Soham Roy

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Summary — Pre-final year CS undergraduate with three accepted publications: ICCVW 2025 on copyright protection for generative AI, and two first-author benchmarks at EMNLP Main and Findings. Research spans agentic AI, benchmarking and evaluation, and retrieval-augmented generation.

Technical Skills

AI/ML: PyTorch • Transformers • Vision-Language Models • Diffusion Models • LoRA • RAG
Agentic Tooling: LangChain • LangGraph • CrewAI • Playwright • Claude Code
Cloud & DevOps: AWS (EC2, S3, Athena) • Docker • Git • Linux
Programming: Python • C++ • SQL • JavaScript
Databases: Snowflake • MongoDB • PostgreSQL • Redis
Development: FastAPI • React • Flask • REST APIs

Work Experience

Cisco **May 2026 – July 2026**
Software Engineering Intern, Network Product Engineering *Bangalore, India*

- Built CVP Insights, an internal platform unifying telemetry from five Cisco product lines (Catalyst Center, Switching, Routing, SD-WAN, Wireless) into one Snowflake warehouse, queryable in natural language (Snowflake Cortex, FastAPI, React)
- Designed a multi-tier entity-resolution pipeline reconciling 16,000+ inconsistently named customer accounts across 13 tables, auto-resolving 76% of escalation records and routing low-confidence matches to human review
- Built the platform's RAG layer over Cisco documentation and an LLM batch-classification pipeline that reclassified 3,400+ mis-tagged accounts; validated with a 50-case eval suite of positive and adversarial queries

Virtual Foundry **June 2025 – September 2025**
AI and Backend Developer Intern *Remote*

- Built EVA, an end-to-end automated hiring pipeline: FastAPI backend with 80+ REST API endpoints, MongoDB Atlas, Google OAuth/JWT authentication, and Docker-based AWS deployment
- Integrated AI-powered resume screening via Google Gemini VLM and automated voice interviews through VAPI.ai, processing thousands of candidate applications
- Implemented logging and monitoring with Railway and AWS CloudWatch across Twilio and OpenAI integrations

RespAI Lab **June 2024 – Present**
Research Member

- Published at ICCVW 2025: dynamic inference-time framework for copyright shielding in diffusion-based image generation, requiring no model retraining (arXiv:2503.16171)
- Conducting ongoing research in machine unlearning, agentic AI safety, and benchmark development, with two first-author papers accepted at EMNLP

Education

Kalinga Institute of Industrial Technology (KIIT DU) **2023 – 2027**
B.Tech in Computer Science and Engineering
CGPA: 9.34/10.00 *Expected: May 2027*

Key Projects

GeoAgent: A Benchmark for Active Visual Geolocation [GitHub] **Accepted, EMNLP Findings**
In collaboration with IIT Bhubaneswar

- Built an agentic scaffold on frontier Vision-Language Models enabling embodied navigation of real-world Street View environments for active geolocation, instead of static image analysis
- Developed a 1,200-location benchmark across 100 cities evaluating VLM spatial reasoning and multi-step navigation, with human baselines and ablations revealing geographic bias across developed and developing regions

SCAMMER4U: Benchmarking PII Leakage in Autonomous Web Agents [arXiv] **Accepted, EMNLP Main**
In collaboration with BITS Pilani and Lam Research

- First-author, pre-registered benchmark measuring whether web agents leak users' personal information when manipulated by adversarial websites during routine tasks
- Designed 91 attacker-controlled environments paired with benign controls under an 8-axis taxonomy of manipulation tactics, jointly measuring PII leakage and attack detection across 4 frontier models

Position of Responsibility

Microsoft Learn Student Ambassadors: KIIT Chapter **Jan 2024 – Present**
Lead: AI/ML Team

- Lead the AI/ML team, handling project planning and technical direction; ran InferenceX, a hands-on workshop on LLM fundamentals and inference. Engaged with 4000+ participants.
- Maintain an open-source Stable Diffusion LoRA fine-tuning framework with 10+ contributors (GitHub)

Achievements

- **All India Rank 82** in Amazon ML Challenge, 2025; selected for **Amazon ML Summer School**, 2025
- Finalist at HackNITR 5.0, National Institute of Technology Rourkela's premier hackathon
- First Place, Microsoft Learn Student Ambassadors Internal Hackathon (2024, 2025)
- Department Rank 1 in 5th semester with SGPA 10.00/10.00